

How Deep Does Interoception Go?

A research summary — prepared 2026-09-27

What interoception is

Interoception is the nervous system's ongoing sense of the body's own internal state — heartbeat, breathing, hunger and fullness, temperature, pain, and gut sensations. It runs constantly and mostly below conscious awareness, surfacing into perception only for certain signals and only for certain people more than others.

How accuracy is measured, and how much it varies

Researchers mostly measure interoceptive accuracy with heartbeat counting or heartbeat discrimination tasks — asking someone to track or match their own heartbeat without taking their pulse. Individual differences are large: some people can barely tell their own heartbeat from an external rhythm, while others are quite precise. Interoception also isn't a single skill — perceiving a resting internal state is a different measure from perceiving a changing one (e.g. during an emotional or autonomic spike), so "how good someone is at interoception" depends on which dimension is being tested.

How far down the signal actually goes

The real anatomical pathways are: vagal afferents, spinal afferents, enteric-nervous-system routes, and humoral/immune signaling originating largely from the gut. These converge on the dorsal vagal complex in the brainstem, a hub that regulates heart rate, breathing, hunger, satiety and digestion, and relays onward to the hypothalamus, insular cortex, and anterior cingulate cortex — the regions where interoceptive signals actually become conscious awareness. Notably, this is still an active, unsettled research frontier: in September–October 2025 a \$14.2 million project launched specifically to map the body's "hidden sixth sense," underscoring that the full depth and resolution of this system isn't mapped yet.

Can it be trained or extended?

Yes, to a real degree. Interoception-based interventions — HRV biofeedback, mindfulness, and structured interoceptive training — show evidence of improving emotional regulation and autonomic self-awareness. The most striking data point is from trained-group studies of the Wim Hof Method: trained participants showed significantly elevated epinephrine and dramatically suppressed pro-inflammatory cytokines (IL-6, IL-8, TNF-alpha) under an induced immune challenge — real, measurable influence over autonomic and immune output through breathing and concentration training. The often-cited single-subject "Iceman" case can't be generalized on its own, but the trained-group results are considerably more solid evidence that the ceiling on conscious influence over "involuntary" systems is higher than assumed, and moves with practice.

Why it matters clinically

Interoceptive dysfunction — either blunted or distorted perception of internal state — is linked to anxiety, depression, addiction, alexithymia (difficulty identifying one's own emotions), obesity, and diabetes. This is a large part of why the field is drawing renewed research investment: it sits upstream of several major categories of mental and physical health conditions.

Bottom line: how deep does it actually go?

For most people, conscious interoception plateaus at fairly coarse signals — heartbeat, breath, hunger, gut discomfort — carried by a handful of well-characterized pathways. Fine-grained awareness of specific organ function or immune activity isn't something anyone has by default. But there is real evidence that training (biofeedback, breathwork, meditation) shifts how much of the normally sub-threshold signal becomes consciously accessible, and can measurably change autonomic and immune output, not just perception of it. The actual ceiling isn't well mapped yet — which is exactly why a multi-million-dollar project to map this system just started.

Sources

The heart rate discrimination task (psychophysical method) —
<https://www.biorxiv.org/content/10.1101/2021.02.18.431871.full.pdf>

Sensitivity to changes in rate of heartbeats as a measure of interoceptive ability —
<https://journals.physiology.org/doi/full/10.1152/jn.00059.2021>

Can You Feel Your Heartbeat? The Answer Says a Lot About You (Vice) —
<https://www.vice.com/en/article/connection-between-heartbeat-anxiety/>

The body sends a signal: Perspectives on interoception (PLOS Biology) —
<https://journals.plos.org/plosbiology/article?id=10.1371/journal.pbio.3003517>

Intestinal interoception: A nexus of environment-body-brain interactions (Neuron) —
[https://www.cell.com/neuron/abstract/S0896-6273\(25\)00923-7](https://www.cell.com/neuron/abstract/S0896-6273(25)00923-7)

Scientists launch \$14.2 million project to map the body's "hidden sixth sense" (ScienceDaily) —
<https://www.sciencedaily.com/releases/2025/10/251025084616.htm>

Interoception (Wikipedia overview) — <https://en.wikipedia.org/wiki/Interoception>

Research on 'Iceman' Wim Hof suggests it may be possible to influence autonomic nervous system and immune response (ScienceDaily) — <https://www.sciencedaily.com/releases/2011/04/110422090203.htm>

The effectiveness of the Wim Hof method on cardiac autonomic function... (PMC) —
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10579249/>